HW6 Part 1 Rubric

Video Questions (1pt each)
Please record a short (3-5 minute) video of yourself answering the following questions about your code.

1. Explain in your own words the difference between the commands “git add” and “git commit”
2. In what situation might a red black tree be preferred over a plain binary search tree? Why?
3. Suppose there is red black tree, which already contains a black root node with the value 18 and a red node on the left side with the value 12. If you call `Insert(7)` on this tree, what area of your code will be executed? Carefully walk through your code and give the line numbers, that will be executed to complete an `Insert(7)` operation on this tree. If you have them, you do not have to explain the helper functions `RotateRight()` or `RotateLeft()` line by line.
4. What is the benefit of using `#define COLOR_RED 0` and `#define COLOR_BLACK 1` in this code base?
5. The RBTNode is described in the video lecture. Where in your code-base did you implement the RBTNode and why did you choose to put it there?
6. You will earn one point if your video is appropriate length (3-5 minutes). You don’t need to record any response to this “question.”

Your video should include your own voice / face, and the sections of code you’re referring to while answering the questions.

Code / Program Assessment Rubric (1pt each)
Please make sure your program correctly addresses the following points.

1. **Compiles and Runs** Valid C++ program with a Makefile that compiles and runs (the student’s tests) without crashing.

2. **Tests** Code passes “group #1” tests
    constructor, insert 1 node, toInfixstring, toPrefixstring

3. **Tests** Code passes “group #2” tests
    insert 2 nodes, insert 3 nodes

4. **Tests** Code passes “group #3” tests
    insert 4+ nodes, edge-cases, large trees, copy constructor, contains, getmin, getmax